

Supplementary Material: PAR-VAE

Auditing Physics-Based Discriminability in CT Classification

This document provides extended methodological detail, data integrity results, and training diagnostics to support the main paper. Sections are cross-referenced from the main text where relevant.

1 9-Stage Data Integrity Protocol

We applied the following automated checks to both cohorts before any model training. All checks passed unless otherwise noted.

Check 1 — File Integrity. Verified physical existence and readability of all generated NumPy arrays. Result: 0 missing files (100% integrity across both cohorts).

Check 2 — HU Range Verification. Audited global Hounsfield Unit statistics to confirm the $[-1000, 400]$ HU clipping operated correctly. Result: global mean $HU = -614.9 \pm 79.1$, consistent with air-filled lung parenchyma. Zero extreme values outside the clipping range were detected.

Check 3 — Mask Integrity. Flagged slices with mask area below 1,000 pixels as non-diagnostic. Result: 0 slices failed this criterion after central-quartile slice extraction. Our Middle-30 strategy successfully excluded non-informative apical and basal slices without any additional filtering step.

Check 4 — Slice Sampling Consistency. Audited slices extracted per patient volume. Result: Cohort A mean 21.1 slices/patient (median 22); Cohort B mean 29.3 slices/patient (median 30). Higher yield in Cohort B reflects more distributed pathology in moderate disease.

Check 5 — Physics Feature Validation. Compared HU distributions between Normal and COVID cohorts. Result: Normal mean $HU = -368.5$; Mild COVID mean $HU = -338.8$ ($\Delta \approx 30$ HU), consistent with GGO density increase. COVID std dev 391.2 vs Normal 405.6 reflects homogenisation from GGOs.

Check 6 — Outlier Detection (IQR Method). Applied Inter-Quartile Range method to all 14 physics features. Result: outlier rates of 3.93% (Mean HU), 3.69% (HU Std Dev), 1.76% (Gradient Mean), and 0.36% (Mask Area). All below 4%, consistent with genuine anatomical variation (patient size, lung inflation) rather than scanner artefact.

Check 7 — Image Quality Audit. Mean gradient magnitude used as sharpness proxy. Dynamic threshold at the 5th percentile flagged potential low-quality scans. Result: 275 slices flagged (5.00%), exactly the expected 5th-percentile rejection rate. No systematic quality bias detected.

Check 8 — Split Balance and Leakage Prevention. Chi-square test applied to label distributions across Train, Validation, and Test splits. Volume-level

splitting was enforced: all slices from a patient volume were confined to a single split. Result: χ^2 p -value = 0.521 (no significant label imbalance). No patient volume was found in more than one split.

Check 9 — Severity Gradient Verification. Mann-Whitney U tests on all 14 features. Result: All features show $p < 0.0001$ between COVID and Normal. Moderate disease (S2) shows larger shifts than mild disease (S1), consistent with the severity gradient hypothesis.

2 Full Dataset Statistics

Table 1. MosMedData severity distribution and class imbalance.

Class Description		Cases	%
S0	No pneumonia	254	22.8
S1	Mild ($\leq 25\%$)	684	61.6
S2	Moderate (25–50%)	125	11.3
S3	Severe (50–75%)	45	4.1
S4	Critical ($\geq 75\%$)	2	0.2
Total		1,110	100.0

Table 2. Dataset splits and class distribution (volume-level splitting).

Split	Cohort A		Cohort B		Cohort C	
	Total	COVID	Total	COVID	Total	COVID
Train	3,878	1,928	1,166	550	1,219	605
Validation	804	398	249	147	264	132
Test	818	424	285	153	277	143
Total	5,500	2,750	1,700	850	1,760	880

Table 3. Physics feature quantitative comparison by severity level.

Feature	Normal	Mild	Moderate	Severe
Mean HU (HU)	−368.5	−338.8	−318.4	−305.8
HU Std Dev (HU)	405.7	391.2	370.5	369.3
Gradient Mean	392.0	373.4	361.2	389.7
Mask Area (pixels)	138,351	148,460	143,405	149,871

3 Training Dynamics

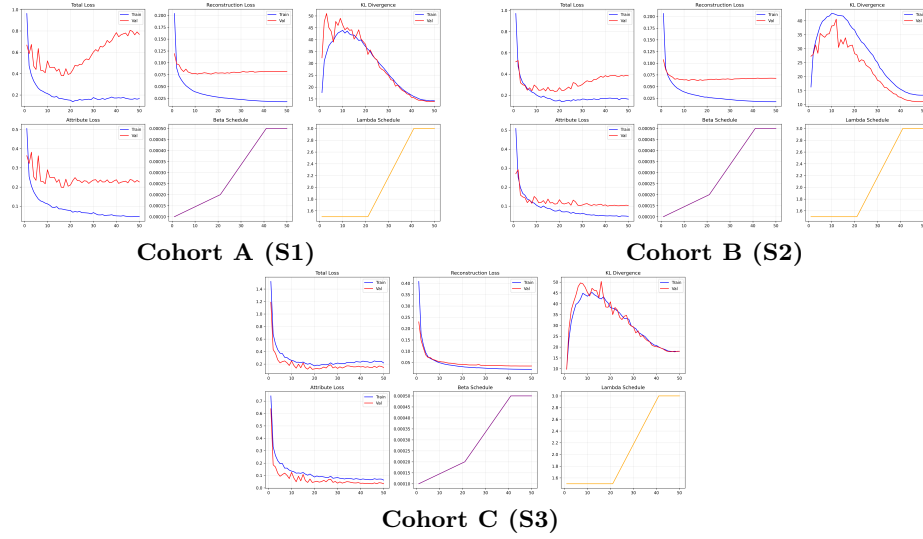


Fig. 1. Training dynamics over 50 epochs: total loss, reconstruction, KL divergence, and attribute loss. KL stabilizes at ≈ 17.5 ; attribute loss converges to ≈ 0.042 on validation.

4 Feature Overlap Analysis

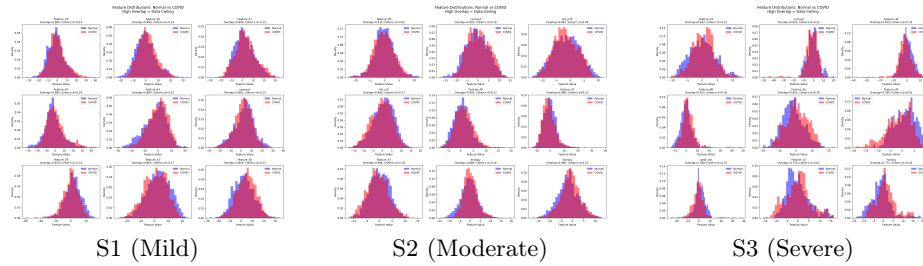


Fig. 2. HU distributions by severity: S1 OVL 0.845 ± 0.080 , S2 OVL 0.841 ± 0.060 , S3 OVL 0.772 ± 0.050 .

Overlap decreases from S1 to S3 as disease severity increases. Physics and learned feature statistics are nearly identical, confirming the ceiling is data-intrinsic. No classifier using univariate features alone can achieve better separation at mild disease levels.

Table 4. Class overlap and separability (mean $\pm$ std, 3 seeds).

Task	Feature Type	Mean OVL	Cohen’s d
S1 vs S0 Physics (14)		0.845 ± 0.080	0.380 ± 0.210
	Learned (top 15)	0.845 ± 0.070	0.410 ± 0.190
S2 vs S0 Physics (14)		0.841 ± 0.060	0.420 ± 0.150
	Learned (top 15)	0.841 ± 0.090	0.440 ± 0.180
S3 vs S0 Physics (14)		0.776 ± 0.050	0.380 ± 0.150
	Learned (top 15)	0.783 ± 0.014	0.368 ± 0.170

5 Classification Results

Table 5. Full classification results across severity levels (mean $\pm$ std, 3 seeds).

Task	Model	Val Acc	Test Acc	Test F1	Test AUC
S1 (Mild)	PAR-VAE	62.0 ± 2.1	62.6 ± 2.8	65.8 ± 2.6	67.4 ± 1.4
	CNN	66.8 ± 1.5	65.2 ± 0.9	68.2 ± 2.5	69.8 ± 1.1
S2 (Moderate)	PAR-VAE	70.8 ± 1.2	66.5 ± 1.6	69.3 ± 2.0	74.6 ± 0.8
	CNN	68.3 ± 2.4	64.1 ± 2.1	66.7 ± 1.9	70.0 ± 1.5
S3 (Severe)	PAR-VAE	98.5 ± 1.1	97.3 ± 2.9	96.7 ± 3.8	99.3 ± 1.0
	CNN	57.1 ± 6.7	60.6 ± 6.3	57.2 ± 11.6	66.0 ± 7.5

6 Feature Coefficients

Logistic regression coefficients (mean $\pm$ std, 3 seeds) for the 14-dimensional physics feature classifier on each cohort, sorted by mean absolute magnitude. Features are detailed in Tables 6–8. High seed-to-seed variance indicates no single feature dominates classification; instability is geometric rather than a symptom of overfitting, especially evident at S3 where all three seed solutions achieve $> 95\%$ test accuracy despite high coefficient variance.

Table 6. Physics feature logistic regression coefficients: S1 vs S0.

Feature	Mean Coeff $\pm$ Std	Mean Coeff
Gradient Std. Dev	-0.517 ± 0.768	0.803
Texture Entropy	-0.086 ± 0.764	0.734
HU Std. Dev	$+0.108 \pm 0.744$	0.714
Homogeneity	$+0.651 \pm 0.459$	0.651
Fractional Occupancy	$+0.283 \pm 0.656$	0.523
HU 25th Percentile	$+0.074 \pm 0.552$	0.461
Gradient Mean	-0.454 ± 0.205	0.454
HU 50th Percentile	$+0.217 \pm 0.526$	0.423
Mean HU	$+0.384 \pm 0.063$	0.384
HU 75th Percentile	$+0.331 \pm 0.270$	0.342
HU 90th Percentile	-0.326 ± 0.256	0.339
Mask Area	$+0.143 \pm 0.305$	0.335
HU 10th Percentile	$+0.040 \pm 0.334$	0.280
GLCM Contrast	$+0.081 \pm 0.197$	0.206

Table 7. Physics feature logistic regression coefficients: S2 vs S0.

Feature	Mean Coeff $\pm$ Std	Mean Coeff
HU 90th Percentile	-0.251 ± 1.114	1.133
Fractional Occupancy	$+0.701 \pm 0.279$	0.701
Mean HU	-0.309 ± 0.692	0.666
Gradient Std. Dev	$+0.046 \pm 0.599$	0.545
HU 50th Percentile	$+0.336 \pm 0.558$	0.544
HU Std. Dev	-0.383 ± 0.630	0.530
HU 25th Percentile	$+0.066 \pm 0.445$	0.434
GLCM Contrast	$+0.147 \pm 0.454$	0.406
Mask Area	-0.142 ± 0.392	0.401
Texture Entropy	$+0.396 \pm 0.237$	0.396
Gradient Mean	$+0.146 \pm 0.451$	0.367
HU 75th Percentile	$+0.238 \pm 0.355$	0.340
HU 10th Percentile	-0.305 ± 0.231	0.319
Homogeneity	-0.009 ± 0.227	0.210

Table 8. Physics feature logistic regression coefficients: S3 vs S0.

Feature	Mean Coeff $\pm$ Std	Mean Coeff
Fractional Occupancy	$+0.401 \pm 1.387$	1.423
Mask Area	-0.049 ± 1.410	1.257
Texture Entropy	-0.042 ± 1.417	1.189
HU 75th Percentile	$+0.684 \pm 0.901$	1.067
HU 10th Percentile	-0.527 ± 0.891	0.978
HU 90th Percentile	-0.001 ± 0.886	0.835
HU 50th Percentile	$+0.827 \pm 0.492$	0.827
Homogeneity	-0.104 ± 0.797	0.689
Gradient Std. Dev	$+0.685 \pm 0.917$	0.685
Gradient Mean	-0.235 ± 0.574	0.615
HU 25th Percentile	$+0.067 \pm 0.610$	0.519
Mean HU	-0.476 ± 0.311	0.476
HU Std. Dev	$+0.114 \pm 0.492$	0.426
GLCM Contrast	-0.103 ± 0.237	0.189